Grade 8 Maths & Science — PDF Stress Test
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Grade 8 Maths & Science — Worksheet Stress Test
20-page document for testing PDF extraction, formula parsing, tables, diagrams, and worksheet structure.
Student: _____ Date: _____
Notation checklist
Read and recognize these characters exactly: ¾ 2½ × ÷ ± >= pi sqrt ° ->
Powers and chemical subscripts: x² H2O CO2 3.5 × 10^4
This packet mixes worked examples, answer tables, independent practice, a diagram, and data interpretation.

Page 2 — Formula reference
Speed = Distance ÷ Time
A = l × b
v = u + at
Symbols: v = final velocity; u = initial velocity; a = acceleration; t = time. For the area formula, l is length and b is b
Practice: write the formula you would use to calculate (a) speed, (b) rectangle area, (c) final velocity when acceler

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Page 3 — Superscripts, subscripts & scientific notation
Algebra: x^2 means x multiplied by itself. If x = 5, then $x^2 = 25$.
Science notation: H_2O is the formula for water; CO_2 is carbon dioxide.
Scientific notation: 3.5 × 10^4 = 35,000.
Practice: evaluate 4^2; write the number represented by 2.1×10^3; identify the subscript in H_2O and CO_2.

Page 4 — Fractions and mixed numbers
Fraction symbols: ¾ = three quarters. Mixed number: 2½ = two and one half.
Practice sums (show working):
1. ¾ + ¼ = _____
2. 2½ - ½ = _____
3. ¾ of 20 = _____
4. Write 1.25 as a fraction in simplest form. _____
Check your notation: keep the fraction bar or fraction glyph clear in your final answer.

Page 5 — Units and measurement
Speed unit: 5 km/h means five kilometres per hour.
Temperature: 25 °C means twenty-five degrees Celsius.
Practice: convert 5 km/h to m/s (use 1 km/h = 5/18 m/s). Convert 25 °C to kelvin using K = °C + 273.
Write units with every numerical answer.

Page 6 — Mathematical symbols
± means plus or minus; >= means greater than or equal to; pi is the circle constant; sqrt denotes a square root.
Practice: choose the correct symbol: (a) 7 __ 5 (>= or ±); (b) sqrt49 = ____; (c) circumference of a circle = ____ ×

Page 7 — Worked example: speed
Example 1: A car travels 120 km in 2 h. Find its speed.
Solution: Speed = 120 ÷ 2 = 60 km/h.
Why? Speed is distance divided by time. The units are km ÷ h, so the result is km/h.
Try: A cyclist travels 18 km in 1.5 h. Find the average speed.

Page 8 — Practice: speed and distance
Solve and show each step.
1. A bus travels 150 km in 3 h. Find speed.
2. A runner covers 400 m in 80 s. Find speed in m/s.
3. A car moves at 60 km/h for 2.5 h. Find distance.
4. A train covers 90 km at 45 km/h. Find travel time.
Answers are not supplied on this page.

Page 9 — Answered practice table

Question	Answer	Working
A vehicle travels 100 km in 2 h. Find speed.	50 km/h	100 ÷ 2 = 50
A cyclist travels 30 km in 3 h. Find speed.	10 km/h	30 ÷ 3 = 10
A car travels at 40 km/h for 2 h. Find distance.	80 km	40 × 2 = 80
A journey of 120 km takes 4 h. Find speed.	30 km/h	120 ÷ 4 = 30

Page 10 — Rectangle area
Formula: $A = l \times b$
Worked example: $l = 8$ cm and $b = 3$ cm. $A = 8 \times 3 = 24$ cm².
Practice (no answers here):
1. $l = 9$ cm, $b = 4$ cm. Find A.
2. $l = 12$ m, $b = 5$ m. Find A.
3. $A = 42$ cm² and $l = 7$ cm. Find b.

Page 11 — Motion formula
For constant acceleration: v = u + at
Example: u = 4 m/s, a = 2 m/s², t = 3 s. Then v = 4 + (2 × 3) = 10 m/s.
Practice: (1) u = 5 m/s, a = 3 m/s², t = 4 s. Find v. (2) u = 0, a = 2 m/s², t = 6 s. Find v.

Page 12 — Mixed calculation practice
Show working and include units.
1. ¾ of 32 = ____
2. sqrt81 = ____
3. 3.5 × 10^4 = ____
4. 5 km/h expressed in m/s = ____
5. 25 °C expressed in K = ____
6. A rectangle is 6 cm by 2½ cm. Find its area.
No answer key is printed for these questions.

Page 13 — Key terms

KEY TERMS

Force: a push or pull on an object.

Speed: distance travelled per unit time.

Acceleration: change in velocity per unit time.

Use each term in a sentence of your own.

Page 14 — Labelled circuit diagram
Study the simple circuit. Trace the wire path and identify the cell, bulb, and connecting wire.
Cell / battery
Bulb
Connecting wire
Closed path
Current direction ->
Schematic (not to scale)
Questions: (1) What component supplies electrical energy? (2) What happens if the circuit path is broken?

Page 15 — Data table: time and distance
A toy car moves along a straight track. Recorded values:
Time (s) | Distance (m)
0 | 0
1 | 2
2 | 4
3 | 6
4 | 8
5 | 10
Read the graph: If distance is plotted on the vertical axis and time on the horizontal axis, what does the straight-li

Page 16 — Graph reading practice
Use the Time | Distance table on page 15.
1. What distance has the car travelled after 4 s?
2. At what time does it reach 8 m?
3. Describe the relationship between time and distance.
4. Calculate the gradient (distance ÷ time) using the 5 s data.
Sketch a distance–time graph on graph paper. Label both axes and units.

Page 17 — Chemical equation
Equation: 2H2 + O2 -> 2H2O
Read the subscripts and coefficients carefully: the small 2 belongs to the formula; the leading 2 multiplies the who
Practice: In 2H2, how many hydrogen atoms are represented? In O2, how many oxygen atoms are represented?

Page 18 — Mixed answer table

Question	Answer	Working
Evaluate x^2 when x = 7.	49	7 × 7 = 49
Find the area when l = 5 cm, b = 4 cm.	20 cm^2	5 × 4 = 20
Find speed for 75 km in 1.5 h.	50 km/h	75 ÷ 1.5 = 50
Find v for u = 3 m/s, a = 2 m/s^2, t = 5 s.	13 m/s	3 + (2 × 5) = 13

Check that units and superscripts remain legible in the extracted answer.

Page 19 — Independent mixed worksheet
Solve without looking back at worked examples.
1. Write the speed formula.
2. Calculate speed: 84 km in 2 h.
3. Calculate area: 7½ cm × 4 cm.
4. Evaluate sqrt121.
5. Write 0.00035 in scientific notation.
6. State what the arrow means in a chemical equation.
7. Define force.
8. From the table on page 15, find distance at 3 s.
Answers intentionally omitted.

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Page 20 — Symbol and extraction checklist
Confirm the PDF reader preserves these exactly:
¾ 2½ 5 km/h 25 °C 3.5 × 10^4 ± >= pi sqrt x² H2O CO2
Formula strings:
Speed = Distance ÷ Time
A = l × b
v = u + at
2H2 + O2 -> 2H2O
Document feature checklist: worked solution; unanswered practice; Question | Answer | Working table; boxed key
End of 20-page stress-test worksheet.